

SMART INDIA HACKATHON 2026



TITLE PAGE

Problem Statement ID – [SIH26103](#)

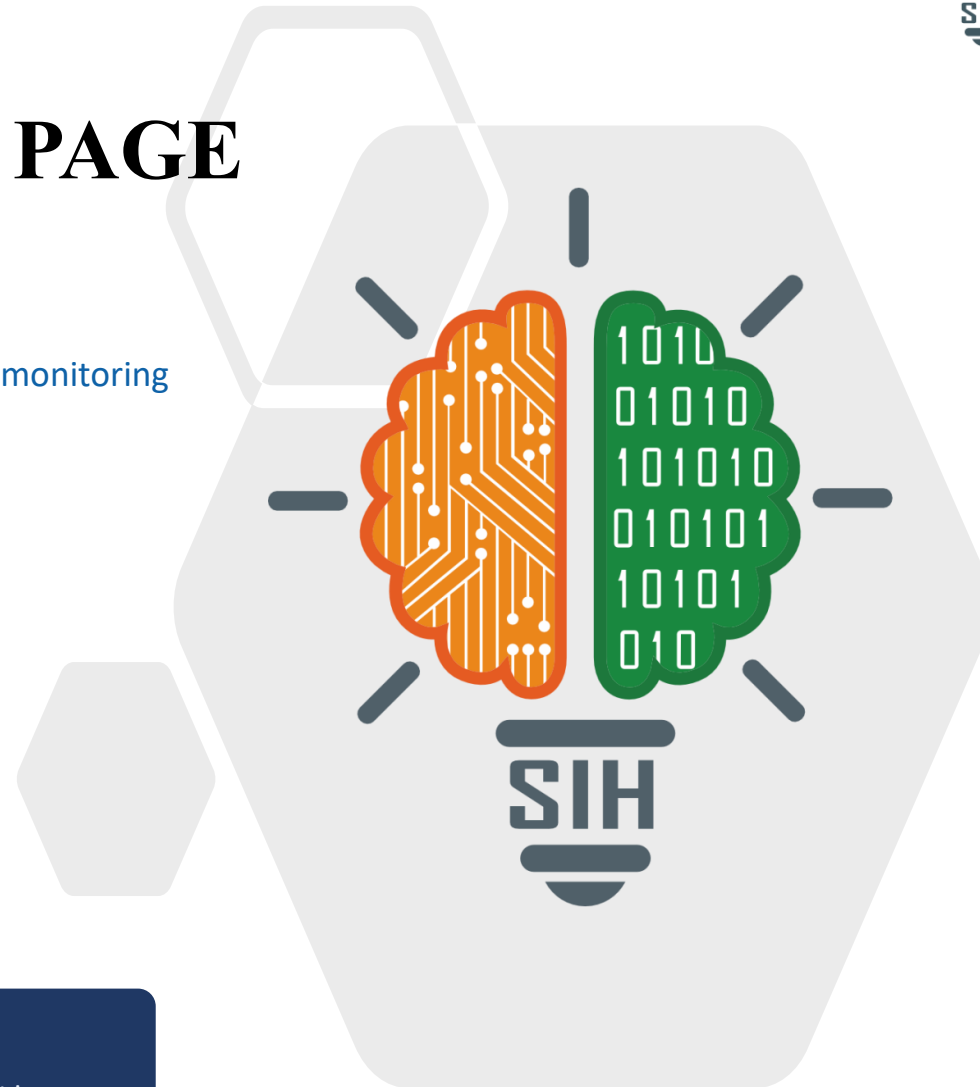
Problem Statement Title- [Predictive analytics for infrastructure project monitoring \(MoSPI / PAIMANA\)](#)

Theme- [Smart Automation / Data Analytics](#)

PS Category- [Software](#)

Team ID- _____

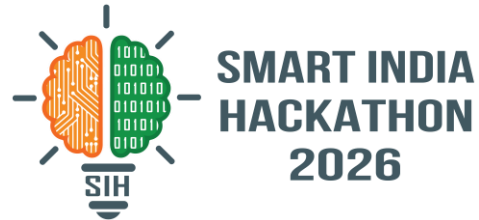
Team Name (Registered on portal)- _____



ANUMAAN — अनुमान

We audit the completion date the agency itself published — and forecast the slip before it is announced.

ANUMAAN — FORECAST AUDITOR FOR PAIMANA



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One-line solution: a project × month panel built from 307 MoSPI Flash Reports that scores every ongoing project each month with a calibrated slip probability, a P10/P50/P90 completion window, the reason, and the authority who can clear it.

Proposed Solution (Describe your Idea/Solution/Prototype)

- Harvest all 307 monthly Flash Reports (FY2001-02 to FY2026-27), each pinned by SHA-256.
- Parse, reconcile against the report's own totals, then build one row per project × month.
- **Four targets, point-in-time features only:**
 - T1 slip ≥ 6 / 12 months (calibrated probability)
 - T2 months-to-commissioning (hazard model, censored)
 - T3 final cost ratio P10/P50/P90 (conformal)
 - T4 months past original date
- Every alert carries a SHAP reason and the owning authority, from a 13-cause taxonomy in MoSPI's own vocabulary.

Detailed explanation of the proposed solution · How it addresses the problem

PAIMANA publishes status. Nobody audits the estimate.

- Jul 2026 report: 1,775 projects, 17 ministries.
- Original ₹33,70,138 cr → revised ₹37,10,642 cr
= ₹3,40,504 cr overrun (+10.10%).
- Apr 2024 report: 792 of 1,838 projects delayed = 43.1%; time overrun spans 1 to 261 months.
- The stated date is revised month after month, and the drift is visible in the panel long before the revision is printed.
- **ANUMAAN scores that drift monthly and publishes what the next revision will most likely say.**

Innovation and uniqueness of the solution

- A panel, not a snapshot: 307 reports, not one month of scraping.
- **A forecast-vs-outcome scorecard per agency — we grade the forecaster, not only the project.**
- Walk-forward validation, reliability diagram and Brier score; conformal coverage measured on held-out months.
- Censoring handled properly: ongoing projects are modelled, never dropped.
- A data-quality gate (DQ001-DQ012) that fails closed, plus identity resolution across OCMS / PAIMANA / PMGID codes.
- 4 distinct report layouts (2019-2026) already parsed and reconciled.

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TECHNICAL APPROACH



Technologies to be used (e.g. programming languages, frameworks, hardware)

Ingest and parse

- Python 3.11 · requests · pdfplumber, one parser per layout · SHA-256 manifest

Store and compute

- DuckDB for the panel and features · pandas · pandera schema checks

Model

- LightGBM plus a regularised logistic baseline — the PS asks for that comparison
- lifelines for the discrete-time hazard · MAPIE for conformal intervals · SHAP for reasons

Serve

- FastAPI and a React dashboard · model card and datasheet with every release

Deploy

- NIC MeghRaj VM, 4 vCPU / 16 GB · logs held in Indian jurisdiction · GIGW-aligned UI

Methodology and process for implementation (Flow Charts/Images/ working prototype)

1 HARVEST

307 reports via the portal report endpoints; manifest + SHA-256

2 PARSE

per-layout parsers: OCMS · Part I/II · Synopsis+Tables · PAIMANA

3 RECONCILE

detail rows against the report's own summary totals — fail closed

4 RESOLVE ID

exact code, then fuzzy name within ministry and cost band, then manual residual

5 PANEL

one row per project × month; point-in-time features only

6 MODEL

walk-forward folds with a label embargo; logistic vs LightGBM

7 ALERT

calibrated probability + P10/P50/P90 + reason + authority

2.4 pages/sec

measured parse throughput (572 pages in 239 s)

307 / 93

reports listed · reports from FY2019-20 onward

~1.6 lakh

project-month rows in the FY2019-20 panel

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FEASIBILITY AND VIABILITY



Analysis of the feasibility of the idea

Technical — measured, not assumed

- Data reachable: 307 reports, HTTP 200 application/pdf verified 16 Sep 2026.
- One-time corpus parse ≈ 8.9 h (76,750 pages $\div$ 2.4 pages/s); monthly top-up ≈ 64 s.
- Storage ≈ 2.5 GB of PDFs and under 200 MB of DuckDB; LightGBM trains in seconds on 4 vCPU.

Economic — ₹2.0 lakh per year to run

- MeghRaj VM ₹1.44 L + 100 GB storage ₹0.06 L + ₹0.5 L contingency.
- No licence cost: every component is open source.

Potential challenges and risks

- Layout drift: 4 distinct report formats between 2019 and 2026, and a single parser silently returns zero rows.
- **Panel breaks: the Oct 2025 report covers 820 projects and excludes MoRTH, against 1,775 in Jul 2026.**
- Sentinel and missing values: 01/1900 dates, NA approvals, 0% progress with crores already spent.
- Right-censoring: most projects are still running, so plain regression on finished ones is biased.
- Time leakage: an aggregate computed over the full history inflates every score.

Strategies · for overcoming these challenges

- One parser per era, plus reconciliation against each report's own totals; a parse that disagrees is rejected, never used.
- **Panel-break flag: a ministry leaving the report is recorded as censoring, not completion.**
- DQ001-DQ012 gate: BLOCKER rows are never scored and appear as “insufficient data” with a reason code.
- A discrete-time hazard on the project-month table keeps ongoing projects in.
- Walk-forward folds with a label embargo, and an as-of feature audit before any number is shown.

PROOF — every figure below is read from that month's own Flash Report (₹ crore)

Report	Projects	Original cost	Revised / Anticipated	Overrun	%
Jul 2019 (OCMS)	1,623	19,33,390.22	23,21,502.84	3,88,112.62	20.07%
Jun 2025 (pre-PAIMANA)	1,595	26,85,568.07	29,75,429.26	2,89,861.19	10.79%
Jul 2026 (PAIMANA)	1,775	33,70,138	37,10,642	3,40,504	10.10%

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IMPACT AND BENEFITS



₹5.65 lakh cr

cost overrun on 1,981 ongoing projects, Apr 2026

43.1%

delayed against original schedule (792 of 1,838, Apr 2024)

₹6.33 cr

idle capital per project for each month of delay

₹500 cr / yr

value of acting 1 month earlier on 10% of delayed projects

Potential impact on the target audience

MoSPI · Infrastructure and Project Monitoring Division

- A ranked monthly watch-list instead of a 153-page PDF.

17 line ministries and implementing agencies

- Each alert names the blocking cause and the authority who can clear it.

PMG / DPIIT and NITI Aayog

- Escalation aimed where the base rate says it pays.

Parliament, CAG and citizens

- Restores the word the reports stopped using: from Sep 2025 PAIMANA reclassified “delayed” projects as “ongoing”.

Benefits of the solution (social, economic, environmental, etc.)

Economic — the calculation

- ₹19,26,100 cr expenditure ÷ 1,775 projects = ₹1,085 cr average capital committed per project.

- **At a 7% cost of public capital: $1,085 \times 0.07 \div 12 = ₹6.33$ cr idle per project-month.**

- 79 projects (10% of the 792 delayed) pulled forward by one month = $79 \times 6.33 = ₹500$ cr a year.

- **Against a ₹2.0 lakh per year run cost, break-even is 0.003 project-months.**

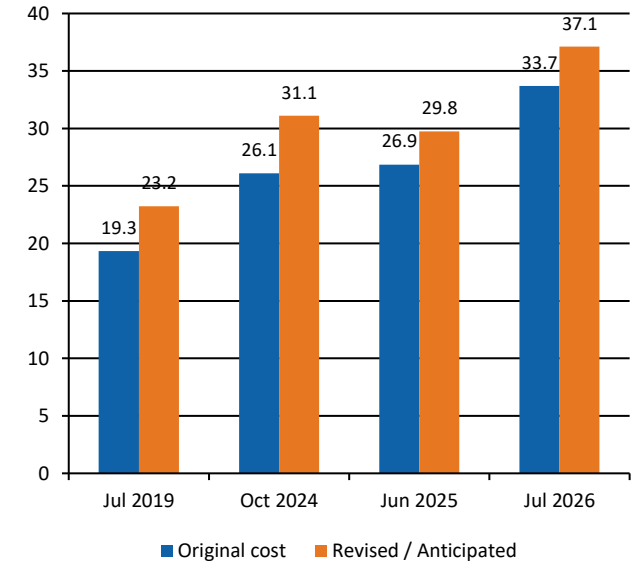
Social and governance

- Calibrated public risk, and an agency-level record of forecast accuracy.

Environmental

- Shorter idle-asset periods, and less rework and re-tendering per completed project.

Original vs revised cost (₹ lakh crore)



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RESEARCH AND REFERENCES



Details / Links of the reference and research work — primary data

MoSPI PAIMANA portal, report archive (opened 16 Sep 2026)

- paimana-proj.mospi.gov.in/ReportPage and [/ReportPage/ArchiveProjectMonitoring](http://ReportPage/ArchiveProjectMonitoring)
- 307 monthly Flash Reports listed, FY2001-02 to FY2026-27; 93 of them from FY2019-20.

Reports read end to end for this submission

- Jul 2019 · Apr 2024 (Part I and II) · Oct 2024 · Jun 2025 · Oct 2025 · Jul 2026
- Each file pinned by SHA-256; every figure on slides 2 to 5 is read from these.

Journalism establishing the transparency gap

- S. Chavaly, “When ‘delayed’ becomes ‘ongoing’”, ThePrint, 13 Aug 2026: 1,981 projects, ₹5.65 trillion overrun, PAIMANA from Sep 2025.

Method and literature (DOIs verified via Crossref, 15 Sep 2026)

- Andrić, Lin, Cheng and Sun (2024). Cost and causes of overruns in South Asian infrastructure. Sustainability 16(24):11159. doi:10.3390/su162411159 — n = 138, 3.3% mean overrun, random forest best.
- Egwim et al. (2021). Applied AI for predicting construction project delay. Machine Learning with Applications 6:100166. doi:10.1016/j.mlwa.2021.100166
- Gneiting, Balabdaoui and Raftery (2007). Probabilistic forecasts, calibration and sharpness. JRSS-B 69(2):243-268.
- Candès, Lei and Ren (2023). Conformalized survival analysis. JRSS-B 85(1):24. arXiv:2103.09763
- Grinsztajn, Oyallon and Varoquaux (2022). Why do tree-based models still outperform deep learning on tabular data? NeurIPS Datasets and Benchmarks.
- Cantarelli, Davis, Pinto and Turner (2025). Reference class forecasting: promises, problems. Production Planning and Control. doi:10.1080/09537287.2025.2578708

Our gap statement: the closest published work uses 138 completed projects. We use about 1.6 lakh project-month rows across 1,700 to 2,000 live projects, and evaluate the way a forecaster must — walk-forward, calibrated, with intervals.

No published benchmark exists for delay prediction on Indian central-sector projects, so we establish the first walk-forward baseline rather than claim to beat a state of the art that does not exist.